Graphical Pressure Mapping of a 2288 Sensing-Point Matrix Pressure Sensor Using Raspberry Pi

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***Abstract*—A pressure sensor reading method has been developed continuously along with the different types of requirements, including in medical fields. A method to read piezoresistive matrix pressure sensors with 2288 sensing points is described in detail. The presentation of the output is in 2-D color graphics representing the pressure with a specific color. The matrix sensor is intended for mapping the body surface pressure of bedridden patients. The pressure sensor points are arranged into a 44 x 52 matrix rows and columns. Selecting the row and column will result in choosing the correct sensing points. The selection of rows and columns utilizes multiplexers and demultiplexers to reduce the input-output pins of the microcontroller. This system requires only one channel analog-to-digital converter (ADC) because the pressure sensor points are selected using a raster scan of row and column matrix. The pitch between sensors (0.5 cm pitch)provides very detailed information about the position and the value of the pressure. Many applications require such precision, such as the profile of foot pressure, the palm of hand pressure, and the back of a bedridden patient. The speed of reading the entire matrix is about 1 second, which is fast enough for this application. An Atmega2560 microcontroller performs the raster scanning process and converts the voltage into digital values. Atmega2560 sends the data serially via USB port to a Raspberry Pi computer. This computer shows the data into a four-color gradation heatmap that maps the pressure to the sensor.**

Keywords—pressure sensor, piezoresistive, heatmap, color gradation, raspberry pi

I. Introduction

Matrix pressure sensors are widely used in industry as robotic skin and biological sensing. There are several types of pressure sensors, such as organic field transistors (OFET) [1], piezoresistance [2], piezoelectricity [3], and capacitors [4]. In this paper, the matrix pressure sensor uses piezo resistance with a pressure range of 0-20 kPa to map the pressure of human body areas on a flat surface. Currently, the test of pressure matrix sensors is on the soles of the feet and palms on hard surfaces.

The principle of the pressure mapping system will be the same for other areas of the body, such as the buttocks area, which can reduce the risk of decubitus [5]. Decubitus occurs due to pressure on one area of the body continuously [6]. The prevention of decubitus in immobilized patients in developing countries is to move the body position of the patient manually by nurses at a specified period, which is still widely in use. Various techniques and technologies continue to improve the quality of patient care at an affordable cost and ease of manufacturing so that it can be used for more patients in need [7]–[11].

The pressure mapping technique will provide accurate information about the critical pressure point. Widening the area surrounding the pressure point will reduce the amount the pressure at one point. This redistribution of pressure is very effective in reducing decubitus. This pressure mapping will be used to verify the distribution of the force exerted on a certain part of the body.

The pressure sensor sheet comprises of 2 pieces of plastic material with thin-film conductors arranged in a matrix [12]–[16]. There are 44 conductors arranged in the horizontal row direction on one sheet, and 52 conductors are arranged in the vertical column direction on the other sheet. The two sensor sheets are stacked together with piezoresistive material in between to act as conductors of the sensing point. The piezoresistive material senses the mechanical pressure and changes the impedance value of the

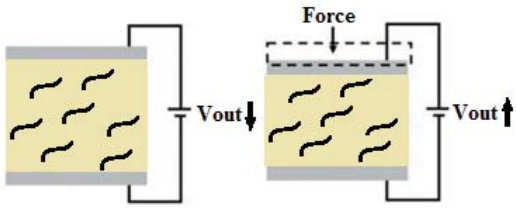

Fig. 1. The illustration of changes at a sensing point on the sensor: (a) when not pressed, (b) when pressed

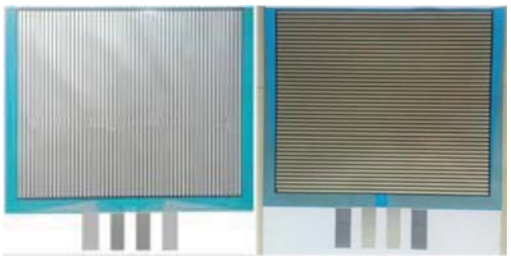

Fig. 2. The matrix pressure sensor

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material, as shown in equation (1) [17].

Fig. 1 shows an overview of the material between contact points located at the two sensor sheets. The piezoresistive material is carbon nanotube [2], [12], [13], [18]. When the material is under mechanical pressure, the distance between the two conductors becomes closer (look at Fig. 1), and the sensor impedance decreases; therefore, the value of the voltage increases. The pressure sensor sheet area is 32 x 37.5 cm2, and the sheet consists of 2288 sensing points arranged in a 44 x 52 matrix. The separation between sensor points (column and row) is 0.5 cm. Fig. 2 shows the pressure sensor sheets on both sites of the sheets. The left picture shows the vertical conductors (consisting of 52 vertical lines), and the right picture shows the horizontal conductors (consisting of 44 horizontal lines). The impedance range of the sensing unit is between 20kΩ and 20MΩ, which correlates with pressure range 0-20kPa.

$$R = \rho L / A \tag{1}$$

II. Method and Fabrication

A. Sensor Controller Design

There are 52 voltage-dividers connected to the vertical lines to read the change of impedance by utilizing the circuits for each sensor point [14], [19]. The higher the pressure on the sensor, the lower the output impedance; hence the measured voltage becomes greater. The entire row side connects to the voltage 52 voltage divider source, and the column side connects to the output voltage. Fig. 3 shows the sensor controller board consists of a 1 to 64 demultiplexer, a 64 to 1 multiplexer, and 52 voltage divider circuits. The analog multiplexer uses 74HC4051 due to its fast turn-on time and turn-off time (69 ns at 5V Vcc working voltage).

Fig. 3 shows the system interconnect among the voltage source, demultiplexer, pressure sensor sheet, voltage divider, multiplexer, and an Atmega2560. The Atmega2560 connects to a Raspberry Pi 4 Model B+ with a 20-inch-monitor via USB interconnect. Atmega2560 controls the 44-line demultiplexer starting from 1- and 52-lines multiplexer to scan the voltage divider. The output of the voltage divider connects to the 10-bit-onboard analog-to-digital converter of Atmega2560.

The digital conversion of the voltage requires 200 µs for every sensing point. With all of the sensing points (52 × 44 = 2288 sensing points) and the overhead, one pressure sensor sheet requires 457.6 ms to complete the pressure reading.

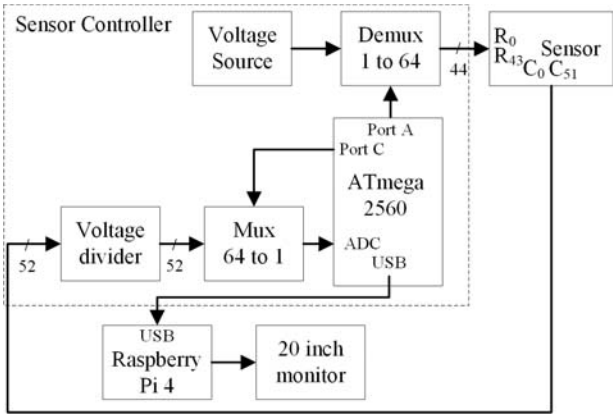

Fig. 3. Diagram block of system

The microcontroller reads all the data of the pressure sensor points and keeps the data in its internal memory. Once the microcontroller finishes reading of the entire sheet, the microcontroller sends the entire data (2288 data at 10-bit length) Raspberry Pi via USB interconnect. Since every byte consists of 8 bits, therefore the 10-bit data must be split into two-byte data chunks and delimited with a comma (",").

Although the data connection is USB interconnect, in this instance, the data is still sent serially using UART (Universal Asynchronous Receiver-Transmitter) at the speed of 115200 bit per second. The protocol sends one data point at a time with a comma (",") as the delimiter character for each data and a slash ("/") to delimit the row. The entire process of data transmission is about 500 ms.

B. Sensor Output Characteristics

The output impedance of the matrix pressure sensor is not linear to the mechanical pressure received [20]. Linearizing the value of the pressure sensor to the voltage requires some data points to the model. Characterizing the output value of this sensor requires six data points.

These sensor output characteristics form a polynomial curve obtained from an experimental method. Every point requires placing lead weights of 5 grams, 55 grams, 105 grams, 205 grams, 305 grams, 505 grams alternately on the sensor in an area of 5x5 mm^2. Applying a simple third-degree polynomial equation yields equation (2). Fig. 4 shows the comparison chart of the sensor output curve and the output curve obtained from the process. A comparison between two curves yields a regression of 0.9992.

$$Y = 2\cdot10^{-7}\, x^3 - 4\cdot10^{-5}\, x^2 + 0.0184\, x - 0.1533 \tag{2}$$

C. Pressure Mapping Design

Fig. 5 shows the four major steps of acquiring the data to the presentation of the data using a heatmap. First, the Atmega2560 converts the analog value into digital using its internal analog to digital converter. It then sends the data serially to the Raspberry Pi in one dimension consisting of 52-data chunks. The Raspberry Pi splits the data into 52-

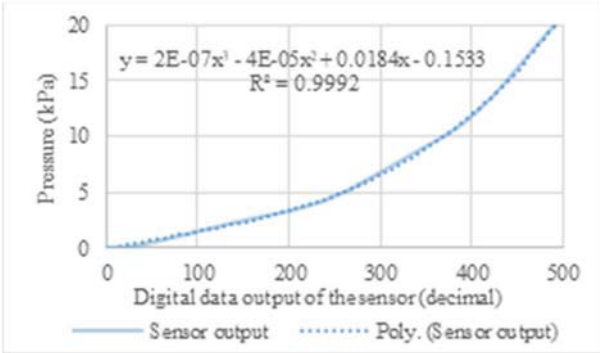

Fig. 4. Graph of sensor properties

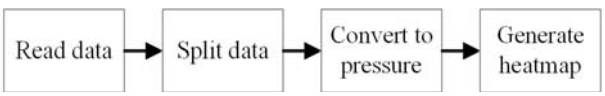

Fig. 5. The system flow process

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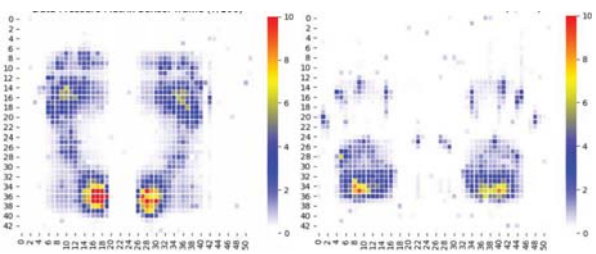
data chunks each and transform the data into a two-dimension data of 52 × 44 data points of voltage. After the conversion into two-dimensions, the Raspberry Pi converts the data into pressure using equation (2).

Fig. 5 shows the flow process inside Raspberry Pi after receiving data from Atmega2560. Raspberry Pi collects the data from the sensor controller is 2 bytes serial data for each sensing point. A data packet received is a collection of data from 2288 sensing points on a matrix pressure sensor. The data processing into heatmap is in four main steps.

Raspberry Pi converts the data to a string data type and then turns the data based on the slash ("/") delimiter. After The separation, there are 44 data rows based on the number of conductors on the matrix pressure sensor in the horizontal position. The data set (52 data sets) are separated based on the comma (",") delimiter into 52 data chunks. The computer then converts the data from the 2-byte data (based on 10-bit data) into a numerical value, and therefore 44 × 52 data are consisting of 10-bit data. Equation (2) converts the 10-bit data of voltage into the value of pressure in kilo Pascal (kPa).

This conversion of data results in a matrix of two-dimensional (2D) data consisting of values of the pressure (in kPa) in 10-bit format. This pressure data is still in the form of numbers in 44 × 52 data. This data is complicated to interpret and then must be converted into a graphical format.

The Matplot library is a library to convert two-dimensional data into heatmap graphics. This two-dimension color representation of values will make the presentation of data easier to visualize [21], [22]. This library is available freely with simple to use high-level commands. Fig. 6a and Fig. 6b show the data conversion from the sensor (already in kPa) using a Raspberry Pi color distribution heatmap (Matplot). The result of the Matplot conversion is a heatmap image in a file with a Portable Network Graphics (PNG) format. There are gradations of four primary colors, namely white, blue, yellow, and red used in this work. The red color represents high pressure and the blue color for the low pressure. Fig. 6 shows an example of a heatmap display. Fig. 6a shows a heatmap of two foot-sole pressed against the matrix pressure sensor, and Fig. 6b shows the palms of two hands pressed against the pressure sensor. The picture clearly shows the location of the high pressure applied to the sheet sensor, which is the end part of the palm and the sole. This result is consistent with the actual condition since the back of the sole, and the back of the palm are the areas with the highest pressure.



(a) (b)

Fig. 6. The heatmap of (a) soles of the feet, (b) palms

III. Experiment and Discussion

A. Testing the Performance of Pressure Conversion Process

This step aims to observe the consistency of output data obtained from one piezoresistive sensor inside the matrix pressure sensor. The test to measure the output of the sensors uses lead scales of 5.6 grams, 56.3 grams, 105.7 grams, 203.5 grams, 303.7 grams, 505.3 grams. Those weights values are calibrated values obtained from a certified scale at the university. Each lead scale is placed at ten sensing points randomly. This process is carried out three times, and then the average value is taken. The maximum and minimum difference between the data resulting from the conversion of the pressure value and the lead mass is recorded.

Table 1 shows the actual weight exerted on an individual sensor at a specific time, and the value of the weight is observed and recorded digitally into the computer for all the six weights mentioned above. The smallest weight shows the most deviation for the minimum and maximum values obtained using the sensor and can be as high as 63% by value, and the least amount of variation is with the weight of higher value such as the 505.3 grams. Table 1 shows all the values of the weight, the deviation of the value from maximum to minimum, the maximum and minimum deviation from the average recorded value. This table shows that the least amount of variation occurs at the highest value, and the maximum is at the lowest range. This indicates that the highest deviation occurs at the lowest weight value, and large weight value generates less deviation

Fig. 7 shows the data trend represented in Table 1. This figure shows that the consistency of the output sensing conversion results in the sensor area is not good enough with the difference between the minimum and maximum conversion results above 25% for masses of 200g and above. The data also shows conversion errors below 20% for the weight of 200g and above, while conversion errors are much higher for smaller weights. Fig. 7 also shows the percentage difference of minimum and maximum values to the average value. This observation shows the data consistency for higher weight values and less consistent for the smaller ones. But smaller weight provides less significant values compare to the larger weight; thus, the error becomes less important.

Table 1. The Pressure Conversion Testing Result

Lead mass (g)	5.6	56.3	105.7	203.5	303.7	505.3			
% difference of (min-avg)	35 %	26 %	28%	12%	15%	14%			
% difference of (max-avg)	43%	16%	23%	17 %	15 %	12 %			
% difference of (min-max)	63%	38%	44%	27 %	28 %	24 %			

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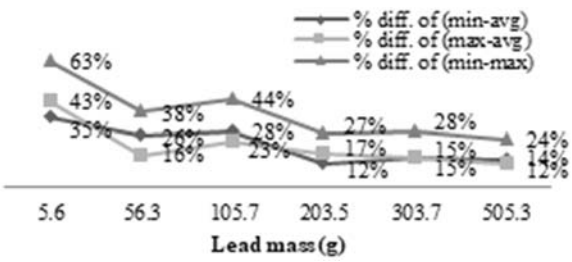

Fig. 7. The percentage differences of minimum, maximum, and average value of the converted pressure data of lead scales on the pressure sensor

The object of the pressure mapping system on this project is mainly the human body. Based on body composition, the lightest body areas that can suppress the sensor are the head and neck, with a mass of 0.07 from body mass [23]. So, the consistency of the output value will be better, and the conversion error will be lower in the intended application.

B. *Testing the Heatmap with Human Subjects*

The heatmap test with human subjects is for the accuracy and consistency of data obtained from the sensors, processed by the Atmega2560, and data to color conversion with the Matplot. For consistency check, the matrix sensor is placed on a hard and flat surface. A subject then presses the matrix sensors using the palm of his hand or soles of his foot. Three participants contribute to this test, with results shown in Fig. 8(a), 8(b), and 8(c). The first participant, whose foot profile is in Fig. 8(a) has a body mass of 55kg with, foot size of 23x9 cm. The second and third participants have a body mass of 47kg, a foot size of 24x9 cm, and a body mass of 90 kg and a foot size 25x11cm have their sole profile in Fig 8(b) and Fig. 8(c).

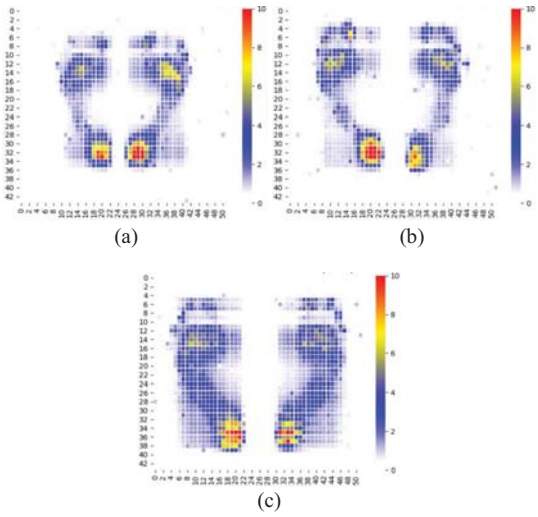

Fig. 8. The heatmap of soles of the feet from 3 subjects, a) person 1, b) person 2, c) person 3

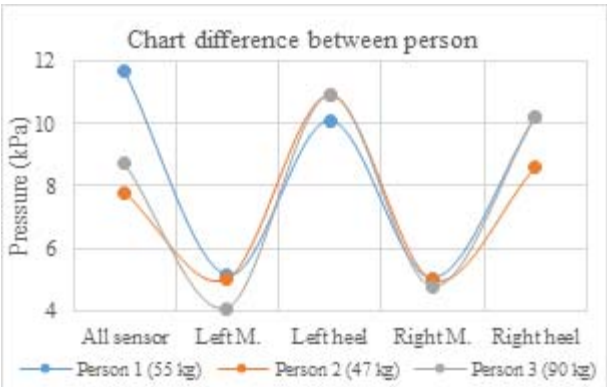

Fig 9. The comparison graph of pressures from 3 subjects

Fig. 8 show the heatmap generated from one of the participants. Fig. 8 shows the resulting heatmap, which displays the pressure map very well. The contours of the soles of the feet and palms appear very clear and distinguishable. The heels show yellow and red color, indicating higher pressure compared to soles (metacarpal). The difference in pressure between the left leg and right leg is also relatively small since the person stands equally between left and right. But the results at this stage still show distortion, especially around areas that are under high pressure, such as the heel area, which is marked by the distribution of light blue.

Fig. 8 shows the heatmaps of soles of the feet from the three participants. It shows that the quality of the heatmap of different subjects is quite similar and part of the soles are recognizable. The area with high and low pressure map is consistent with the contour of the sole for all three subjects. However, the distortion near the region with the highest pressure also consistently appears.

The consistency of heatmap mapping was also tested by comparing pressure data on four areas of body pressure from all three participants. Fig. 9 shows a comparison graph of the average pressure of two regions on each left foot and right foot of the three participants with the average value of pressure obtained from all sensor areas. The observed foot areas are the heel area and the sole (metacarpal). The graph shows a similar trend, high pressure in the heel area, and lower in the metacarpal. The apparent difference can be caused by the position of humans when standing that can lean slightly left or right. Significant differences in chart trends can be the data analysis of the subject's posture.

IV. Conclusion

The method proposed in this paper produces a pressure distribution map in the form of a heatmap with four color gradations. The results of the pressure measurement and conversion have an error of less than 20%. The results of heatmap testing on human subjects showed the system has been able to map the distribution of pressure. The heatmap can show the contour of the soles of the feet and palms quite well. However, this mapping system still has a weakness. It is the presence of distortion around areas that are under high pressure so that the shadow appears on the foot contour. The performance of this mapping system as a whole has been able to map the matrix pressure sensor area that is under pressure and illustrates the magnitude of the pressure with correlated colors.

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